

LAB EXPERIMENT RECORD

SET 8: Employee Performance

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	SATURDAY

SET 8: Employee Performance Analysis Aim

To visualize corporate personnel metrics including operational tenure, cross-department trends, and evaluation outputs using R scripts alongside a dynamic multi-view dashboard inside Tableau.

Dataset: Employee Performance

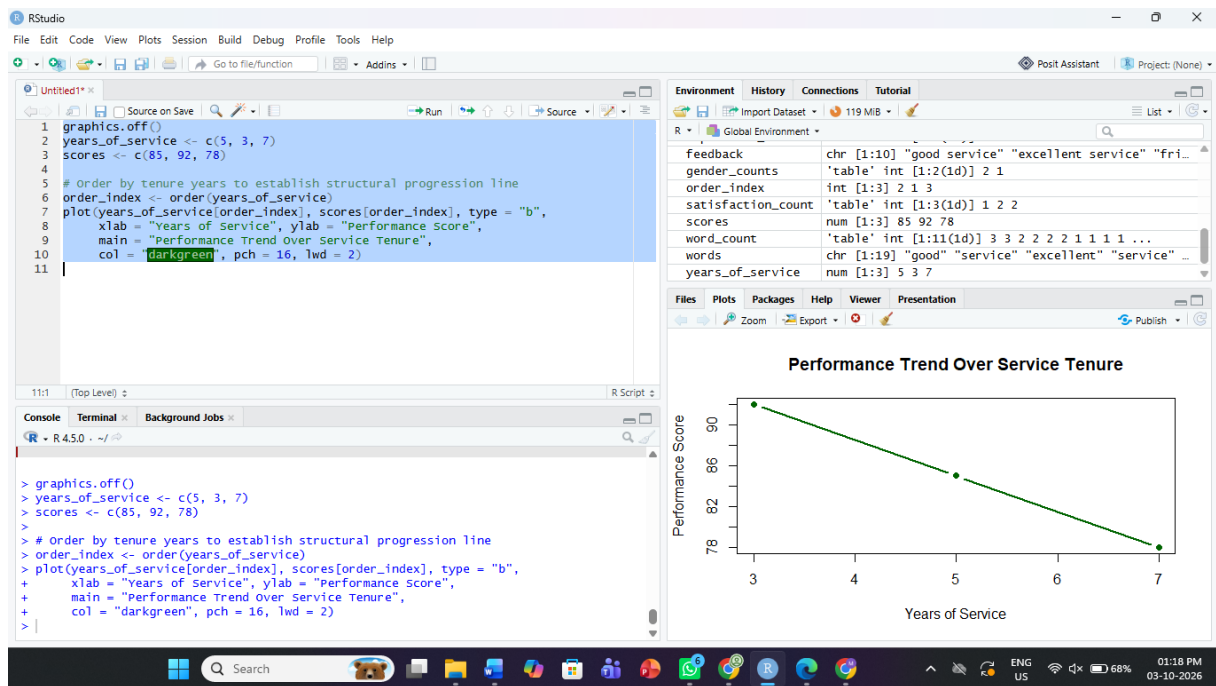
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

PROGRAM 1: Line Chart (Performance over Service Tenure)

Aim: To chart individual employee performance patterns relative to their operational timeline.

```
graphics.off()
years_of_service <- c(5, 3, 7)
scores <- c(85, 92, 78)

# Order by tenure years to establish structural progression line
order_index <- order(years_of_service)
plot(years_of_service[order_index], scores[order_index], type = "b",
      xlab = "Years of Service", ylab = "Performance Score",
      main = "Performance Trend Over Service Tenure",
      col = "darkgreen", pch = 16, lwd = 2)
```



Observation: The trend highlighting score versus service timeline exhibits a downward trajectory among this small sample group as tenure extends.

PROGRAM 2: Bar Chart (Departmental Placement)

Aim: To plot workforce distribution tracking across different functional business units.

```
graphics.off()
```

```
departments <- c("Sales", "HR", "Marketing")
```

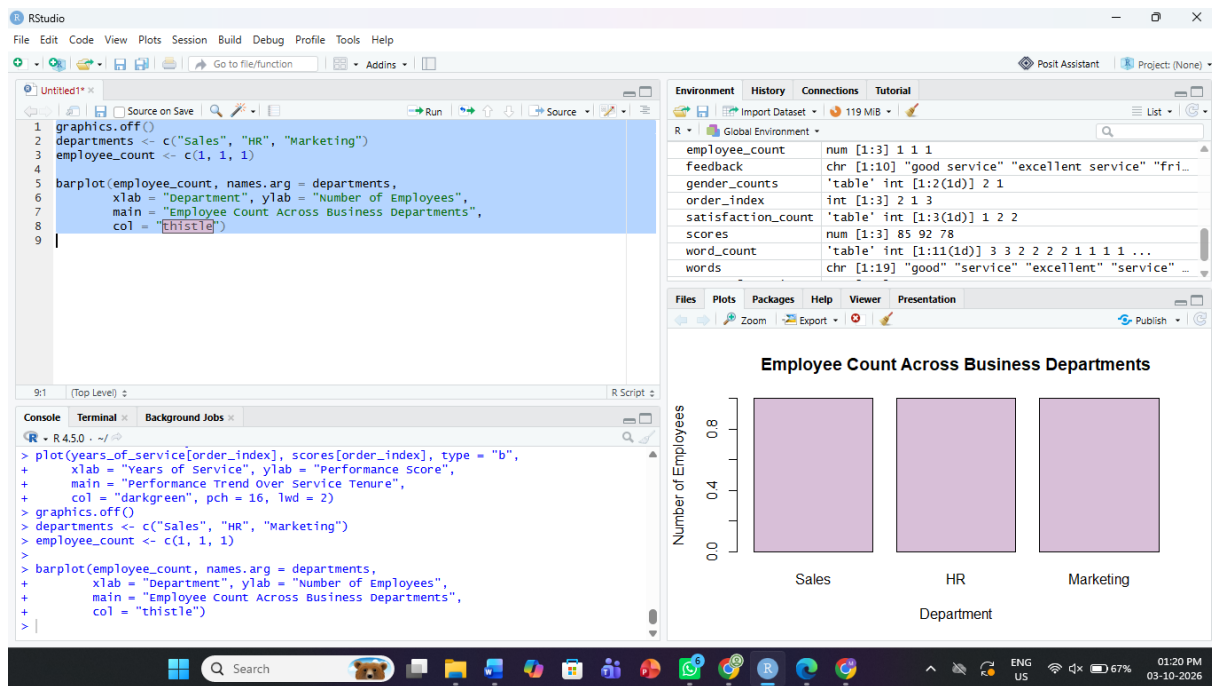
```
employee_count <- c(1, 1, 1)
```

```
barplot(employee_count, names.arg = departments,
```

```
       xlab = "Department", ylab = "Number of Employees",
```

```
       main = "Employee Count Across Business Departments",
```

```
       col = "thistle")
```



Observation: There is a completely equal and uniform distribution of one employee assigned per monitored department.

PROGRAM 3: Performance Performance Summary Matrix Table

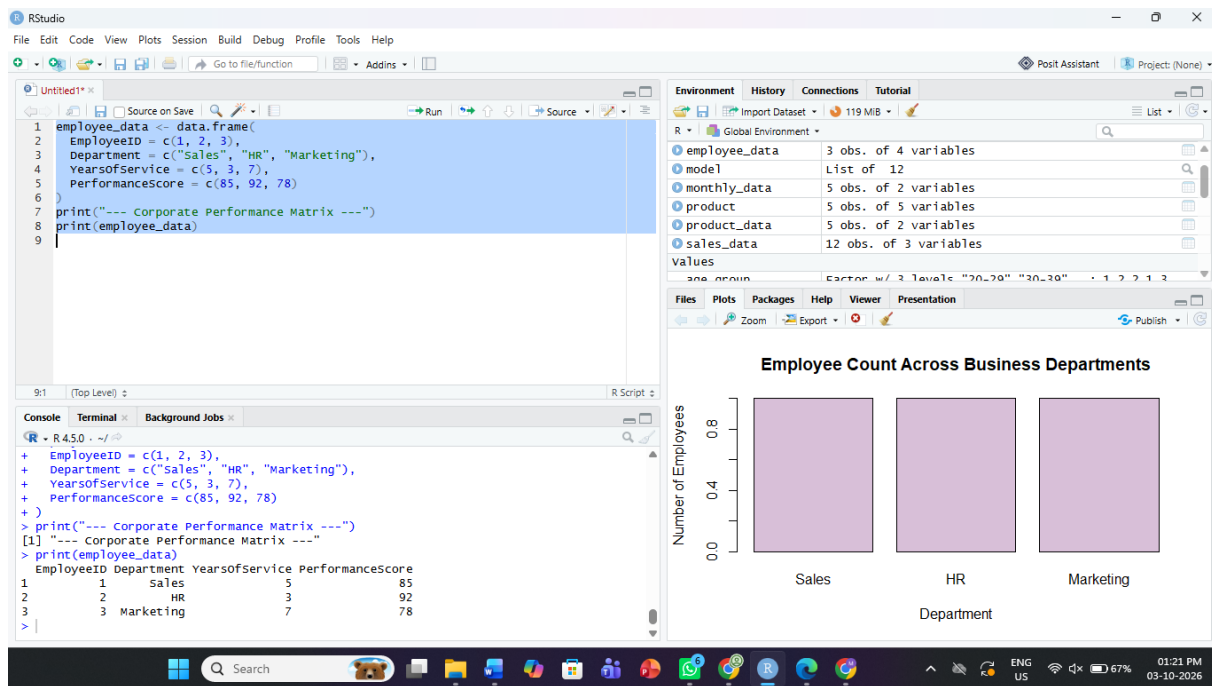
Aim: Build a structured presentation frame summarizing performance evaluation records.

```

employee_data <- data.frame(
  EmployeeID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  YearsOfService = c(5, 3, 7),
  PerformanceScore = c(85, 92, 78)
)

print("--- Corporate Performance Matrix ---")
print(employee_data)

```



Observation: The Human Resources representative holds the highest recorded performance score of 92 points with three years of service.

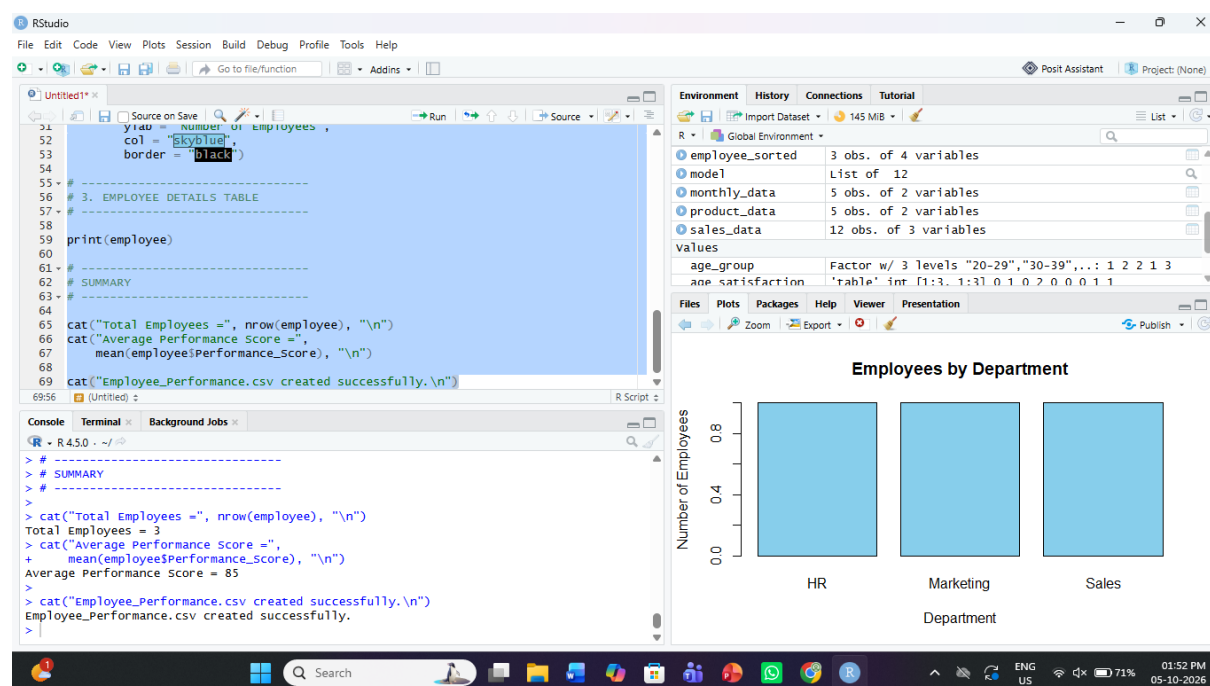
Program 4: Integrated Employee Performance Dashboard in Tableau

Aim: Construct a collaborative Tableau dashboard aggregating charts and operational rosters for interactive personnel analysis.

Steps in Tableau:

1. Save the employee performance dataset matrix as a flat structure file (Employee_Performance.xlsx). Establish external data binding links within the Tableau dashboard interface.
2. **Sheet 1 (Performance Line Chart):** Map Years of Service to Columns (Continuous Dimension) and Performance Score to Rows. Change type selection to Line. Label plot marks with score properties.
3. **Sheet 2 (Departmental Breakdown Bar):** Place Department on Columns and Employee ID (Count) on Rows. Assign distinct colors for department elements via Marks control.
4. **Sheet 3 (Performance Data Sheet):** Drop Employee ID, Department, Years of Service, and Performance Score onto the rows area. Rename to "Detailed Performance Grid".
5. Construct a Master Dashboard workspace layout. Arrange the trend line, department count bar, and reporting matrix side-by-side.
6. Turn on the dashboard interactive control layer using the **Use as Filter** action toggle.

OUTPUT:



Observation: Filtering by a single organizational branch focuses focus areas entirely on specific performance scores and operational timelines of individuals matching that entity context.

Procedure

1. Formulate isolated vector structures containing employee details within the workspace environment.
2. Sort array indices and execute a plot() command using type parameter "b" to trace sequential metrics over time.
3. Map categorical departments into distribution vectors to pass forward to barplot() statements.
4. Structure records inside an automated data layout using R's table generation objects.
5. Create visual worksheets inside Tableau and organize them together onto an integrated presentation canvas.

RESULT:

Employee evaluation datasets were mapped using multi-dimensional visual types in R, with automated filtering control established by using a master Tableau layout.

